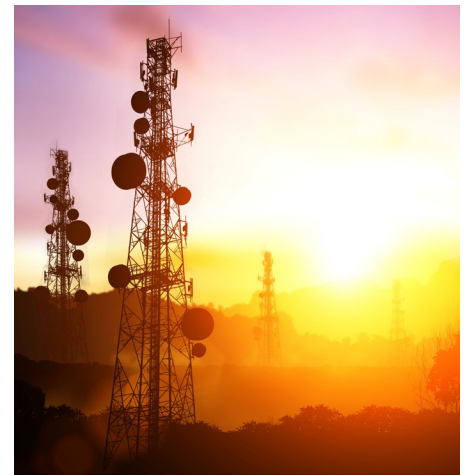


# Model 845-M RF/Microwave Signal Generator



## Features

- Low Phase Noise
- Fast Switching Down to 25  $\mu$ s
- FM, Chirps, Pulse
- Internal OCXO, External Variable Reference
- USB, LAN, GPIB Interfaces

## Applications

- ATE
- LO for frequency converters
- Telecom / SatCom
- System Integration



## Model 845-M

0.01 to 20 GHz Low Noise Synthesizer

# CONTENTS

|                                                                    |           |
|--------------------------------------------------------------------|-----------|
| <b>Contents .....</b>                                              | <b>2</b>  |
| <b>Definitions .....</b>                                           | <b>3</b>  |
| <b>Introduction.....</b>                                           | <b>3</b>  |
| <b>Facts &amp; Figures &amp; Specifications.....</b>               | <b>4</b>  |
| Signal Specifications .....                                        | 4         |
| Frequency Reference .....                                          | 4         |
| Level Performance.....                                             | 4         |
| Reverse Power Protection and VSWR .....                            | 5         |
| Phase Noise .....                                                  | 5         |
| Spectral Purity .....                                              | 5         |
| Modulation Capabilities .....                                      | 6         |
| Pulse Modulation .....                                             | 6         |
| Internal pulse generator.....                                      | 6         |
| Frequency Modulation .....                                         | 7         |
| Phase Modulation .....                                             | 7         |
| Frequency Chirps.....                                              | 7         |
| Sweeping Capability, Sweep type: linear, logarithmic, random ..... | 7         |
| Trigger (TRIG IN): Input is TRIG IN at front panel.....            | 8         |
| Connectors .....                                                   | 9         |
| Front .....                                                        | 9         |
| Rear .....                                                         | 9         |
| <b>Ordering Information .....</b>                                  | <b>10</b> |
| <b>General Characteristics.....</b>                                | <b>10</b> |
| Document History .....                                             | 11        |
| <b>Notes.....</b>                                                  | <b>12</b> |

## DEFINITIONS

- The specifications in the following pages describe the warranted performance of the instrument for  $23 \pm 5$  °C after a 30-minute warm-up period

**Typical:** Expected mean values, not warranted performance

**Min and max:** Parameter range that is guaranteed by product design, and/or production tested. Warranted performance specifications include guard-bands to account for the expected statistical performance distribution, measurement uncertainties, and changes in performance due to environmental conditions.

## INTRODUCTION

### • Ultra-compact, fast and low power consumption Frequency Synthesizer with USB & LAN interface

The Model 845-M is a wideband low phase-noise synthesizer operating from 10 MHz to 20 GHz. The nominal output power is +23 dBm.

The module has a mili-Hz frequency resolution and uses a high-stability internal reference. The internal reference can be phase-locked to a user-settable external reference. For highest phase coherence, multiple 845-Ms can be cascaded with just one master reference clock.

The Model 845-M offers dedicated sweeping capabilities with switching speeds of only 25  $\mu$ s and wideband frequency modulation as well as narrow pulse modulation.

The module has USB and LAN interfaces and can be controlled using the SCPI 1999 command set. Operated with an external 6V DC supply, it consumes less than 10 watts.

## FACTS & FIGURES & SPECIFICATIONS

### Signal Specifications

| PARAMETER         | MIN    | TYPICAL     | MAX    | NOTE                        |
|-------------------|--------|-------------|--------|-----------------------------|
| Frequency Range   | 10 MHz |             | 20 GHz |                             |
| Resolution        |        | 0.001 Hz    |        |                             |
| Phase Resolution  |        | 0.1 deg     |        |                             |
| Switching Speed   |        | 1.5 ms      |        | after SCPI command received |
| CW Mode           |        | 180 $\mu$ s |        |                             |
| Sweep / List Mode |        | 25 $\mu$ s  |        | Option FS                   |

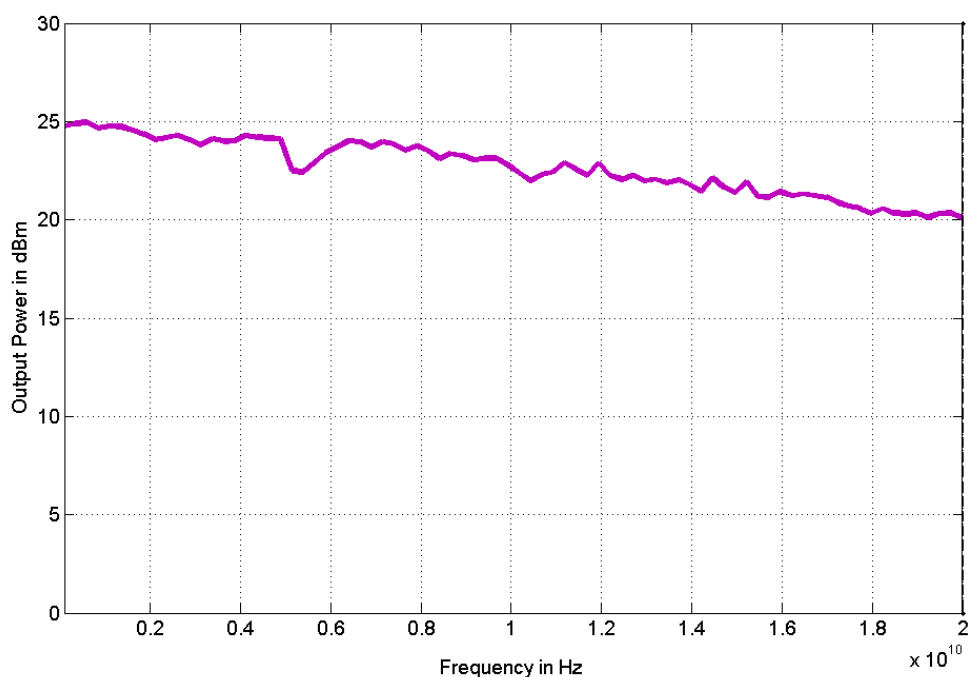
### Frequency Reference

| PARAMETER                           | MIN    | TYPICAL     | MAX           | NOTE                        |
|-------------------------------------|--------|-------------|---------------|-----------------------------|
| Internal reference frequency        |        | 100 MHz     |               |                             |
| Internal Reference Output Frequency |        |             |               |                             |
| Temperature stability               |        |             | $\pm 100$ ppb | 0 to 50 degC                |
| Aging 1st year                      |        | 0.5 ppm     | 1 ppm         |                             |
| Aging per day                       |        |             | 5 ppb         | after 30 days operations    |
| Warm-up time                        |        | 5 min       |               |                             |
| Output of internal reference        |        | 100 MHz     |               |                             |
| Output power                        | 0 dBm  | 5 dBm       |               |                             |
| Output impedance                    |        | 50 Ohms     |               |                             |
| Bypass Internal reference Input     |        | 100 MHz     |               | High phase synchronous mode |
| Phase Lock to External Reference    | 1 MHz  | integer MHz | 250 MHz       |                             |
| Reference input level               | -5 dBm | 0 dBm       | +13 dBm       |                             |
| Reference Bypass Mode               | 5 dBm  |             | +15 dBm       |                             |
| External Reference Lock Range       |        |             |               |                             |
| 1-250 MHz                           |        |             | $\pm 1.0$ ppm |                             |
| Bypass 100 MHz                      |        |             | >100 ppm      |                             |
| Reference input impedance           |        | 50 Ohms     |               |                             |

### Level Performance

| PARAMETER    | MIN    | TYPICAL | MAX    | NOTE       |
|--------------|--------|---------|--------|------------|
| Output power | 18 dBm | 23 dBm  | 26 dBm | (see plot) |

Figure 1: Output Power 0.01 to 20 GHz





## Reverse Power Protection and VSWR

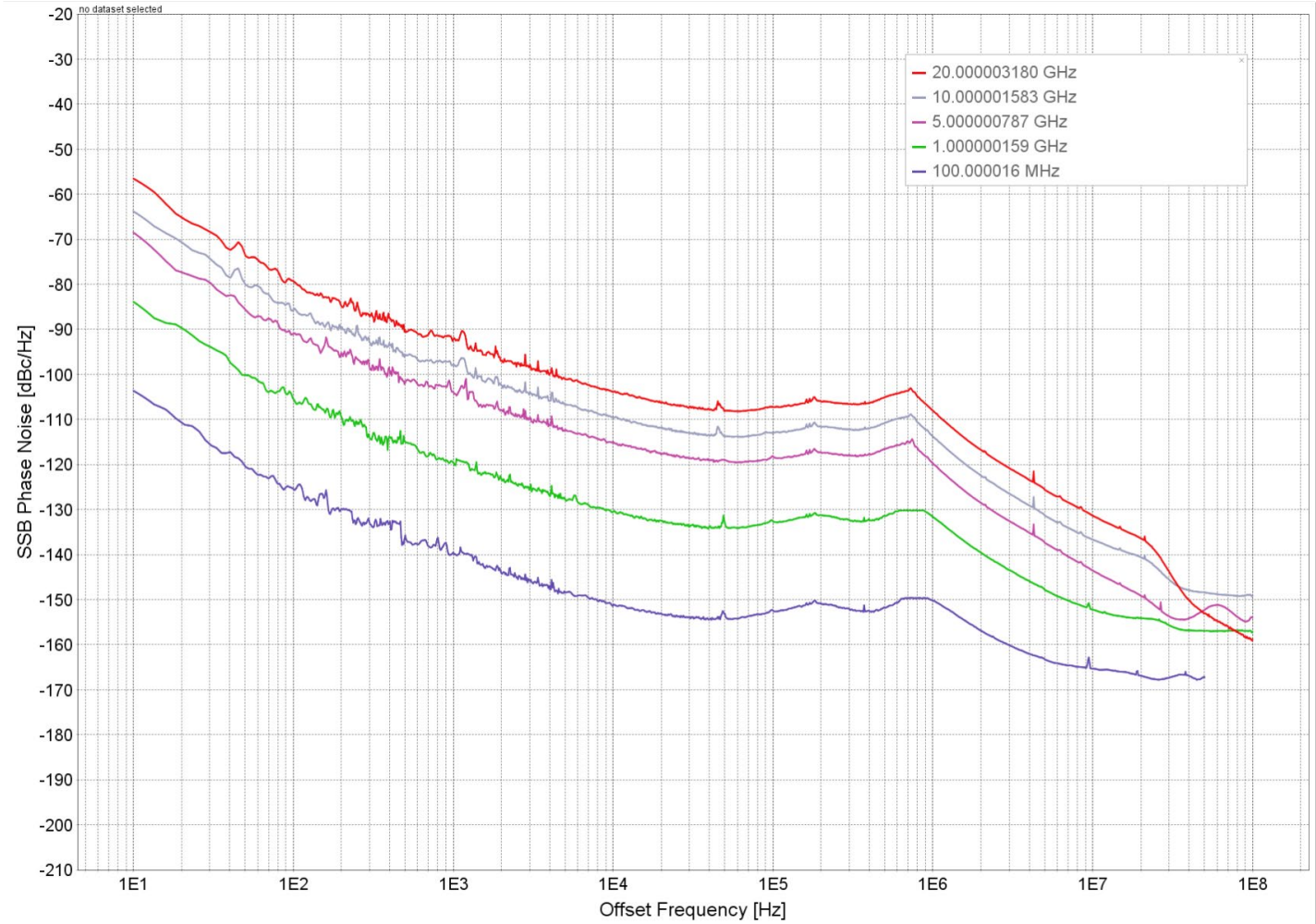
| PARAMETER                       | MIN | TYPICAL | MAX    | NOTE |
|---------------------------------|-----|---------|--------|------|
| <b>Reverse Power Protection</b> |     |         |        |      |
| DC Voltage                      |     | 7 V     |        |      |
| RF Power                        |     |         | 23 dBm |      |
| <b>Output impedance</b>         |     |         |        |      |
| VSWR                            |     | 1.8     |        |      |



## Phase Noise

| PARAMETER                        | MIN | TYPICAL     | MAX | NOTE            |
|----------------------------------|-----|-------------|-----|-----------------|
| <b>SSB Phase noise at 1 GHz</b>  |     |             |     |                 |
| at 1 kHz from carrier            |     | -118 dBc/Hz |     | (see also plot) |
| at 100 kHz from carrier          |     | -128 dBc/Hz |     |                 |
| Wideband noise                   |     | -150 dBc/Hz |     |                 |
| <b>SSB Phase noise at 10 GHz</b> |     |             |     |                 |
| at 1 kHz from carrier            |     | -100 dBc/Hz |     |                 |
| at 100 kHz from carrier          |     | -108 dBc/Hz |     |                 |
| Wideband noise                   |     | -150 dBc/Hz |     |                 |

Figure 2: Phase Noise Performance



## Spectral Purity

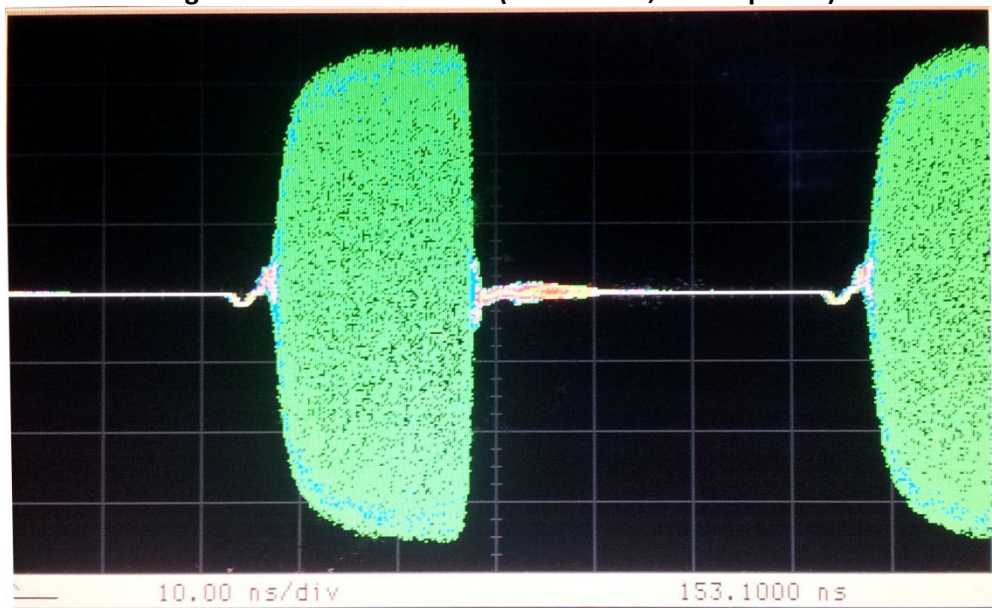
| PARAMETER                    | MIN | TYPICAL | MAX     | NOTE |
|------------------------------|-----|---------|---------|------|
| <b>Spectral purity</b>       |     |         |         |      |
| Output harmonics             |     | -15 dBc |         |      |
| Sub-harmonics                |     | -75 dBc | -60 dBc |      |
| <b>Non-harmonic spurious</b> |     |         |         |      |
|                              |     | -75 dBc | -60 dBc |      |

## Modulation Capabilities

### Pulse Modulation

| PARAMETER                    | MIN    | TYPICAL               | MAX    | NOTE                     |
|------------------------------|--------|-----------------------|--------|--------------------------|
| Modulation source            |        | Internal/<br>External |        |                          |
| Pulse rise/fall time         |        | 7 ns                  |        |                          |
| On/off ratio                 | 30 dB  | 45 dB                 |        | Pout > +10 dBm           |
| Pulse overshoot              |        |                       | 10%    |                          |
| Pulse delay                  |        | 20 ns                 |        |                          |
| Pulse polarity               | 1V     | 2V<br>TTL             |        | AC coupled<br>DC coupled |
| External input threshold     | 0.85 V | 0.9 V                 | 0.95 V | TTL compatible           |
| External input voltage range | -0.5 V |                       | +5.5 V | TTL compatible           |
| External input hysteresis    |        | 60 mV                 |        |                          |

**Figure 3: Pulse Modulation (20 ns width, 100 ns period)**



## Internal pulse generator

| PARAMETER                                   | MIN                        | TYPICAL            | MAX    | NOTE                                    |
|---------------------------------------------|----------------------------|--------------------|--------|-----------------------------------------|
| Repetition frequency (PRF)                  | 0.1 Hz                     |                    | 10 MHz | = 1/T                                   |
| Duty cycle                                  | 1 % to 99 %<br>in 1% steps |                    |        | within specified minimum pulse<br>width |
| Pulse Width settling range                  | 30 ns                      |                    | 1 s    |                                         |
| Pulse Pattern Modulation & Staggered<br>PRF |                            |                    |        | Using internal pattern generator        |
| Programmable pattern length                 | 2                          |                    | 4192   |                                         |
| Pulse width resolution                      |                            | 15 ns              |        |                                         |
| Pulse jitter                                |                            | 2 ns               | 10 ns  |                                         |
| Polarity                                    |                            | Normal,<br>inverse |        | selectable                              |



## Frequency Modulation

| PARAMETER                          | MIN    | TYPICAL                   | MAX     | NOTE                                                                                                                                                                                                                                                                         |
|------------------------------------|--------|---------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modulation source                  |        | Internal                  |         |                                                                                                                                                                                                                                                                              |
| Maximum Frequency deviation (peak) |        | $N \cdot 500 \text{ MHz}$ |         | $< 1.25 \text{ GHz } (N=1)$<br>$1.25 \text{ GHz to } 2.5 \text{ GHz } (N=0.125)$<br>$2.5 \text{ GHz to } 5 \text{ GHz } (N=0.25)$<br>$5 \text{ GHz to } 10 \text{ GHz } (N=0.5)$<br>$10 \text{ GHz to } 20 \text{ GHz } (N=1)$<br>$20 \text{ GHz to } 40 \text{ GHz } (N=2)$ |
| Deviation accuracy                 |        | 0.50%                     | 2%      |                                                                                                                                                                                                                                                                              |
| Distortion (THD)                   |        | $< 1 \%$                  |         | 1 kHz rate, 10 kHz deviation                                                                                                                                                                                                                                                 |
| Modulation rate                    | 0.1 Hz |                           | 800 kHz | 3dB                                                                                                                                                                                                                                                                          |
| Modulation waveforms               | Sine   |                           |         |                                                                                                                                                                                                                                                                              |



## Phase Modulation

| PARAMETER              | MIN    | TYPICAL  | MAX                            | NOTE                                         |
|------------------------|--------|----------|--------------------------------|----------------------------------------------|
| Modulation source      |        | Internal |                                |                                              |
| Phase deviation (peak) | 0      |          | $100 \cdot N \cdot \text{rad}$ |                                              |
| Deviation accuracy     |        | 0.50%    | 2%                             |                                              |
| Modulation rate        | 0.1 Hz |          | 800 kHz                        |                                              |
| Modulation waveforms   |        | Sine     |                                |                                              |
| Distortion (THD)       |        | $< 1\%$  |                                | 1 kHz rate & $N \times \text{rad}$ deviation |



## Frequency Chirps

| PARAMETER               | MIN    | TYPICAL               | MAX                             | NOTE |
|-------------------------|--------|-----------------------|---------------------------------|------|
| <b>Frequency Chirps</b> |        | Linear, ramp, up/down |                                 |      |
| Modulation source       |        | Internal              |                                 |      |
| Bandwidth               |        |                       | 10%                             |      |
| Dwell time              | 1 ns   |                       | 10 ms                           |      |
| Slope                   | 0.1 Hz |                       | $100 \text{ MHz} / \mu\text{s}$ |      |



## Sweeping Capability, Sweep type: linear, logarithmic, random

| PARAMETER                         | MIN                                   | TYPICAL | MAX | NOTE      |
|-----------------------------------|---------------------------------------|---------|-----|-----------|
| <b>Frequency Sweep</b>            |                                       |         |     |           |
| Step time ( $t_{\text{step}}$ )   | 180 $\mu\text{s}$<br>25 $\mu\text{s}$ |         |     | Option FS |
| Dwell time ( $t_{\text{dwell}}$ ) | 15 $\mu\text{s}$                      |         |     |           |



## Trigger (TRIG IN): Input is TRIG IN at front panel

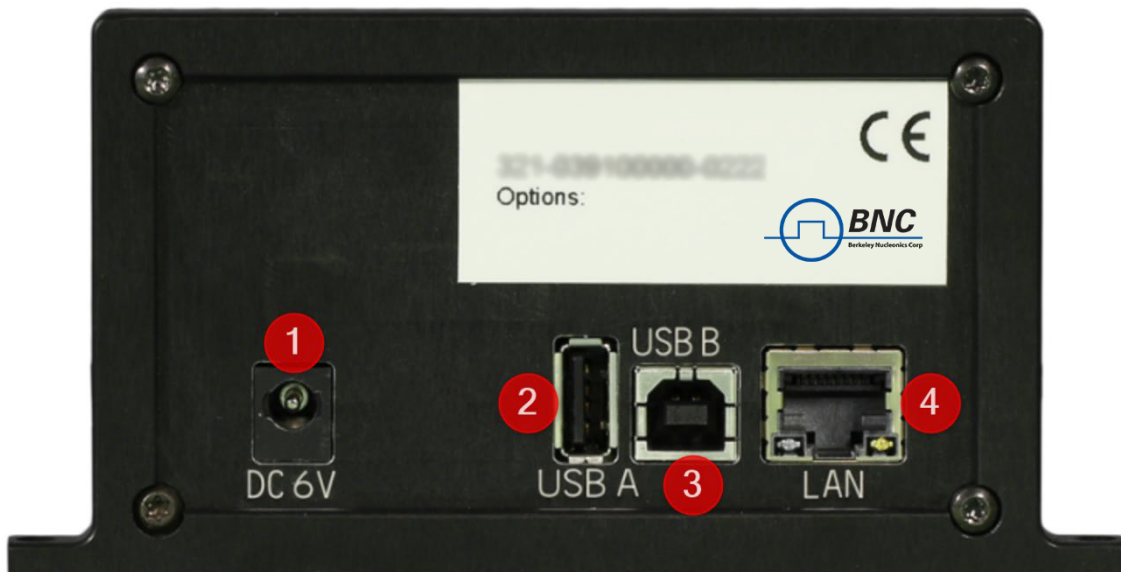
| PARAMETER                            | MIN             | TYPICAL   | MAX    | NOTE                                                |
|--------------------------------------|-----------------|-----------|--------|-----------------------------------------------------|
| Trigger Types                        |                 |           |        | Continuous, single (point), gated, gated direction  |
| Trigger Source                       |                 |           |        | external, bus (LAN, USB)                            |
| Trigger Modes                        |                 |           |        | Continuous free run, trigger and run, reset and run |
| Trigger uncertainty                  |                 | 5 $\mu$ s |        |                                                     |
| External Trigger delay               | 50 $\mu$ s      |           | 40 s   |                                                     |
| External Delay Resolution            |                 | 15 ns     |        |                                                     |
| Trigger Modulo                       | 1               |           | 255    | Execute only on Nth trigger event                   |
| Trigger Polarity                     | Rising, falling |           |        |                                                     |
| External trigger input threshold     | 0.85 V          | 0.9 V     | 0.95 V | TTL compatible                                      |
| External trigger input voltage range | -0.5 V          |           | +5.5 V | TTL compatible                                      |
| External trigger input hysteresis    |                 | 60 mV     |        |                                                     |

Front



1. **Power switch**
2. **RF connector** SMA-type (female)
3. **REF IN** External reference input: BNC female
4. **REF OUT** Internal reference output: BNC female
5. **PULSE** Pulse modulation input: BNC female
6. **TRIG** Trigger input: BNC female

Rear



1. **DC Power plug** (6 V, 3 A)
2. **USB Type A**
3. **USB 2.0** host and device
4. **LAN connection** RJ-45

## ORDERING INFORMATION



| HOST MODEL | PRODUCT     | DESCRIPTION                             |
|------------|-------------|-----------------------------------------|
| 845-M      | 845-M       | 20 GHz Synthesizer, flange-mount module |
| 845-M      | Option 1URM | 19 inch 1HU rackmount enclosure         |
| 845-M      | Option FS   | Ultra fast switching speed              |
| 845-M      | Option GPIB | GPIB interface (only with option 1URM)  |

## GENERAL CHARACTERISTICS

### Remote programming interfaces

Ethernet 100BaseT LAN interface, USB 2.0 host & device, Control language SCPI Version 1999.0

### Power requirements 6V VDC; 20 W maximum

**Mains adapter supplied:** 100-240 VAC in/ 6 V 6.0 A DC out

**Environmental** (Levels similar to MIL-PRF-28800F Class 3/4)

Environmental stress Samples of this product have been type tested to be robust against the environmental stresses of storage, transportation, and end-use; those stresses to temperature, humidity, shock, vibration, altitude, and power line conditions.

**Operating temperature range** 0 to 45 °C

**Storage temperature range** -40 to 70 °C

**Operating and storage altitude** up to 15,000 feet (4600 m)



Safety/EMC complies with applicable Safety and EMC regulations and directives.

**Weight:** ≤ 1.0 kg (2.2 lbs) net

**Dimensions (W x L x H):** 10.5 x 21 x 6 cm [4.13 x 8.27 x 2.36 in]

10.5 x 27 x 6 cm [4.13 x 10.63 x 2.36 in] (with option FS)



**Document History**

| Version/Status | Date       | Author | Notes                                             |
|----------------|------------|--------|---------------------------------------------------|
| V147           | 2019-03-01 | jk     | New layout                                        |
| V148           | 2019-03-27 | mm     | Correction: Frequency Reference Out-put power     |
| V149           | 2021-02-25 | db     | Pulse and trigger input electrical specifications |
| V150           | 2023-10-09 | ee     | Updated layout                                    |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |
|                |            |        |                                                   |

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NOTES

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